

FlowMap Lite v1 Product Specification

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Version: v1.0

Status: Draft (discussion/baseline)

Scope: Process Map + User Journey only

1) Product Summary

Goal

Build a simple, high-utility visual mapping tool that converts raw process/journey documentation into clear, editable maps with:

1. Manual editor (core canvas operations)
2. Intelligent document import (human-reviewable)
3. User journey flow mapping
4. Intelligent layout optimizer

Primary intents (v1 only)

- Process Map
- User Journey Flow

Key product decision

Use ****same project, separate canvases/views**** by default:

- Journey View (stage-based)
- Process View (graph-based)

Allow cross-links between nodes across views.

2) Scope

In Scope

- Intent-first new map flow
- Manual editor for nodes/connectors
- Journey-specific structures (persona, stages, pain points, opportunities)
- Intelligent import wizard with mapping review
- Layout optimization (whole map or selection)
- PNG/PDF export

Out of Scope (v1)

- Real-time multiplayer
- Deep third-party bidirectional sync
- Large template marketplace
- Full BPMN parity

3) Information Architecture

```text

Project

```
|— Canvas: Process View
| |— Nodes/Edges
| |— Layout Profile: Process
|— Canvas: Journey View
| |— Stages + Journey Nodes
| |— Layout Profile: Journey
```





|           |                                                              |
|-----------|--------------------------------------------------------------|
| Persona:  | New Buyer                                                    |
| Stages:   | Awareness   Consideration   Signup   Activation              |
| Actions:  | View ad   Compare plans   Fill form   Complete setup         |
| Touchpts: | Instagram   Pricing page   OTP email   In-app guide          |
| Pain:     | unclear CTA   choice overload   OTP delay   missing guidance |

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### ### 5.6 Optimizer Modal

```

```text
      +-----+
      | Intelligent Layout Optimizer |
      +-----+
      | Scope: (o) Entire map ( ) Selection |
      | Preserve: [x] Pinned nodes [x] Stage columns |
      | Goals: [x] Min crossings [x] Uniform spacing |
      | [Cancel] [Preview] [Apply] |
      +-----+
...
---
```

6) Functional Requirements

6.1 Manual Editor

- Create/edit/delete nodes and connectors
- Inline text editing
- Drag/drop, pan/zoom
- Multi-select align/distribute
- Undo/redo
- Pin node position

6.2 Intent-Specific Modeling

Process Map

- Start/End, Step, Decision
- Directed connectors
- Branch labels optional

User Journey

- Persona (single persona v1)
- Ordered stages
- Actions per stage
- Touchpoints
- Pain points and opportunities

6.3 Intelligent Import

- Inputs: pasted text, txt, md, docx, pdf, csv
- Intent classification + confidence
- Structure extraction and schema mapping
- User mapping review before generation
- Fully editable output

6.4 Intelligent Layout Optimizer

- Optimize all or selection
- Process constraints: flow direction, crossing minimization
- Journey constraints: stage-column preservation, row consistency
- Respect pinned nodes
- Preview/apply/revert

6.5 Export

- PNG/PDF

- Full map or viewport

7) Data Model (v1)

```
``ts
type Project = { id: string; name: string; createdAt: string; updatedAt: string };
type Canvas = {
  id: string;
  projectId: string;
  intent: "process" | "journey";
  name: string;
  createdAt: string;
  updatedAt: string;
};
type Node = {
  id: string;
  canvasId: string;
  type: string;
  label: string;
  x: number;
  y: number;
  width: number;
  height: number;
  pinned?: boolean;
  metadata?: Record<string, string>;
};
type Edge = { id: string; canvasId: string; sourceId: string; targetId: string; label?: string };
type Stage = { id: string; canvasId: string; name: string; order: number };
type CrossLink = {
  id: string;
  projectId: string;
  fromCanvasId: string;
  fromNodeId: string;
  toCanvasId: string;
  toNodeId: string;
};
type ImportRun = {
  id: string;
  projectId: string;
  sourceType: string;
  detectedIntent: "process" | "journey";
  confidence: number;
  warnings: string[];
};
``
```

8) Ticket Breakdown (Frontend / Backend / AI-Parser)

Frontend

- FE-01: Intent-first New Map flow
- FE-02: Core canvas editor
- FE-03: Intent-aware palettes + journey lane UI
- FE-04: Properties panel + pinning
- FE-05: Import wizard UI (3 steps)
- FE-06: Optimizer modal with preview/apply/revert
- FE-07: Same project, dual-view navigation + cross-link jump
- FE-08: Export modal

Backend

```
BE-01 -> FE-01 -> FE-02 -> FE-03/04 -> BE-03+FE-06 -> BE-02
    \\\                                \\\                \\\
    \\\                                -> BE-05+FE-08
        -> AI-01 -> BE-04 -> AI-02 -> AI-03/04 -> AI-05+FE-05 -> AI-06 -> AI-07
```

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11) Risk Notes and Mitigations

1. ****Import quality variance (PDF/DOCX complexity)****
 - Mitigation: normalization + OCR fallback + mandatory mapping review.
2. ****Token/cost variance on long docs****
 - Mitigation: chunking and staged extraction with hard caps.
3. ****Layout disrupts manual intent****
 - Mitigation: pinning + selection-only optimize + preview/revert.
4. ****Journey/process semantic mixing****
 - Mitigation: separate canvases by default, cross-link instead of merge.
5. ****Hallucinated parser edges****
 - Mitigation: confidence surfacing + explicit unresolved/ambiguous queue.
6. ****Undo complexity across import + optimization****
 - Mitigation: revision snapshots per destructive action.
7. ****Latency spikes****
 - Mitigation: async job pipeline + progressive status feedback.
8. ****Regression drift in prompts/models****
 - Mitigation: golden-set evaluations gated in CI.

12) Suggested Runtime Token Policy

- Soft cap/import: 12k tokens
- Hard cap/import: 20k tokens
- If over cap: auto-switch to staged extraction (outline -> drilldown)
- Monitor:
 - tokens/import
 - import confidence
 - manual correction volume
 - latency per stage

13) Acceptance Criteria (MVP)

1. User can create and edit both intent types from blank.
2. User can import supported documents, review mappings, and generate editable maps.
3. Layout optimizer can preview/apply and revert changes safely.
4. User can keep Journey and Process in one project as separate canvases and cross-link nodes.
5. User can export map in PNG/PDF.

14) Download Notes

This file is self-contained and can be downloaded directly as:

`FLOWMAP\_LITE\_V1\_SPEC.md`

For distribution as PDF, render this Markdown via any markdown-to-PDF tool.